

Practical: Star Topology Simulation

Aim

To understand Star Topology and simulate its creation using a Hub and a Switch in Cisco Packet Tracer, and to study the differences between Hub and Switch as networking devices.

Theory

Star Topology:

Star topology is a network structure in which all devices (nodes) are connected to a central device such as a hub or a switch. All communication between devices passes through the central device. If one peripheral device fails, it does not affect the rest of the network; however, if the central device fails, the entire network goes down.

Hub:

A hub is a basic networking device that connects multiple computers in a network. It works at the physical layer of the OSI model. When a hub receives data from one device, it broadcasts the data to all connected devices, regardless of the intended recipient.

Switch:

A switch is an intelligent networking device that operates at the data link layer of the OSI model. It forwards data only to the specific device for which the data is intended by using MAC addresses, making communication faster and more secure.

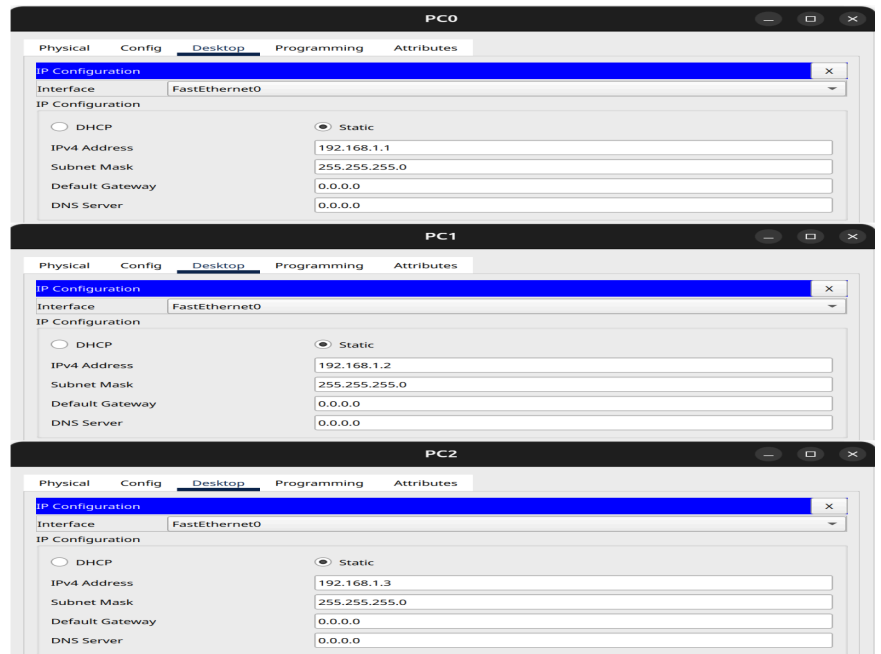
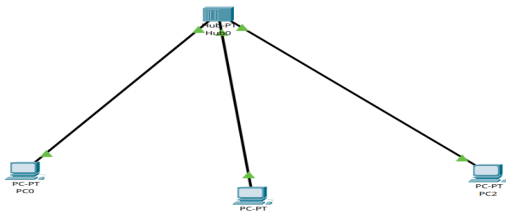
Difference between Hub and Switch

Hub	Switch
Broadcasts data to all devices	Sends data only to the destination device
Works at Physical Layer	Works at Data Link Layer
Less efficient	More efficient
Lower cost	Higher cost
No MAC address table	Uses MAC address table

Simulation 1: Star Topology using Hub

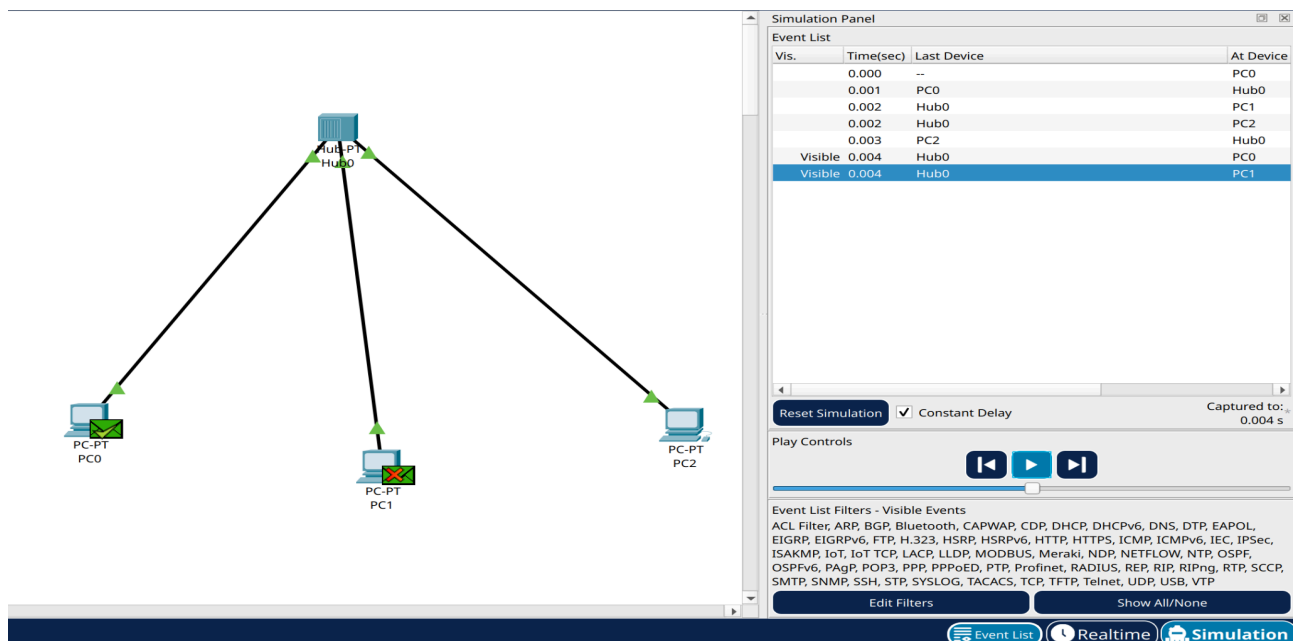
1. Open Cisco Packet Tracer.
2. Drag one Hub from the network devices section into the workspace.
3. Drag multiple PCs (PC0, PC1, PC2, PC3) into the workspace.
4. Use Copper Straight-Through cables to connect each PC to the Hub.
5. Assign IP addresses to all PCs in the same network.
6. Switch to Simulation mode and test communication between PCs.

Screenshots (Hub):



[Screenshot 1: PCs connected to Hub]

[Screenshot 2: IP configuration of PCs]

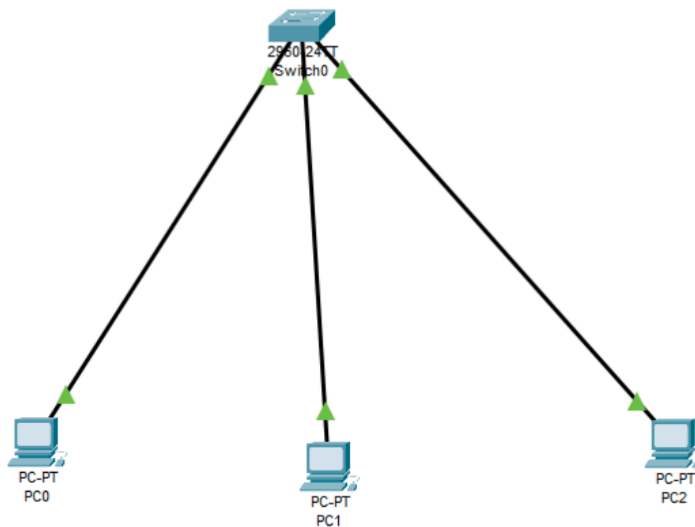


[Screenshot 3: Successful data transmission]

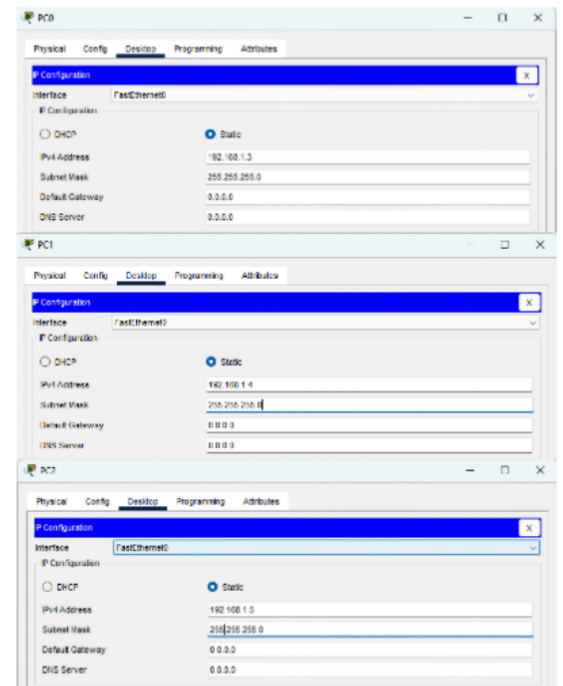
Simulation 2: Star Topology using Switch

1. Open Cisco Packet Tracer.
2. Drag one Switch from the network devices section into the workspace.
3. Drag multiple PCs (PC0, PC1, PC2, PC3) into the workspace.
4. Use Copper Straight-Through cables to connect each PC to the Switch.
5. Assign IP addresses to all PCs in the same network.
6. Switch to Simulation mode and test communication between PCs.

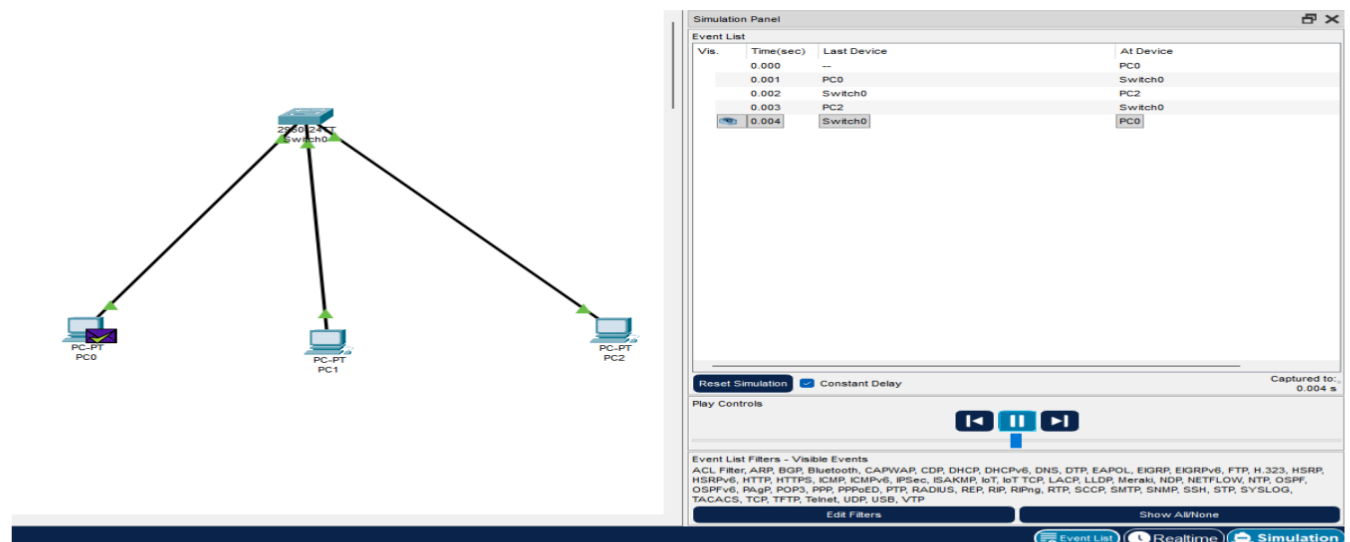
Screenshots (Switch):



[Screenshot 1: PCs connected to Switch]



[Screenshot 2: IP configuration of PCs]



[Screenshot 3: Successful data transmission]